Learning-based Onboard Guidance for Fuel-Optimal Powered Descent

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I. Introduction

During the powered descent phase of a human-Mars mission, a landing vehicle is required to decelerate from a high speed to a near-zero speed and land at a preflight-designated location with high precision by maneuvering a retro-propulsion system. The nonlinear vehicle dynamics, operational constraints, and limited propellant during the mission make the precision planetary landing a challenging task. We aim to generate real-time optimal guidance laws to control the retro-thrust magnitude and direction to achieve precise landing using minimum fuel consumption. The fuel-optimal powered descent guidance (FOPDG) problem has been extensively investigated. Both direct and indirect methods have been developed to solve the FOPDG problems.

The basic idea of direct methods is to approximate an optimal control problem via well-defined collocation methods to convert it into a parameter optimization problem that can be solved by the state-of-the-art nonlinear optimization methods, such as nonlinear programming (NLP) [1, 2]. While efficient for solving a large class of problems in different applications, the convergence of direct methods is dependent on the initially guessed values of the unknown variables and global convergence to a local optimum is not guaranteed in general. NLP has been applied to solve entry, descent, and landing related problems [3, 4], where a good initial guess of the unknown variables is generally required to achieve a convergent result. Due to the nonconvex nature of the vehicle dynamics and operational constraints, many relaxation methods have been developed to reduce the computational complexity. Among them, the convex optimization approach has been frequently used to solve the powered descent guidance problems, where a of convex problems are formulated

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and solved via the interior point method [5–9]. Among them, the lossless convexification has been proposed to solve the FOPDG problem, which has been implemented and flown on-board a real rocket [10, 11]. In addition, a framework combining pseudospectral method and convex optimization has been proposed to solve FOPDG with improved accuracy [12]. However, with the nonlinear dynamic and operational constraints, traditional approaches based on direct methods cannot guarantee global convergence to an optimal solution with rigorous proof in general cases.

Based on Hamiltonian and first-order necessary conditions, indirect methods generally require a good guess of the initial adjoint variables to solve the two-point-boundary-value problem (TPBVP) [13]. However, in most cases, these adjoint variables have no explicit physical meaning, which makes it difficult to guess their initial values within a reasonable scale [14]. Approximation [15–18] and shooting methods [19, 20] have been developed to solve TPBVPs. However, due to the sensitivity of adjoint variables, existing methods cannot guarantee convergence to the exact solution. By ignoring the constraints on state and control variables, the author in [21] has obtained the exact analytical solution of the optimal powered descent guidance problem. Another work in [22] developed a robust optimal control approach for solving FOPDG based on an indirect method with approximate dynamics. Other types of indirect methods can be found in [23–26]. Although the indirect methods have the advantage of obtaining continuous-time solutions, their practical application in optimal control with nonlinear dynamics and constraints is limited.

For optimal control problems with varying/uncertain parameters in the dynamical model or states/controls, supervised learning (SL) has been applied to find optimal control solutions in real-time based on a well-trained neural network [27, 28]. SL has been applied to solve optimal guidance of powered descent problems. For example, [29] applied the deep neural network (DNN) to learn the relationship between initial conditions and optimal state-control pairs. In addition, [30] applied the recurrent neural network to predict the optimal thrust from a sequence of states. However, these learning-based optimal control methods require to generate large-scale artificial neutral networks to calculate all state or control variables, or both of them, to achieve high accuracy of the learning results, which creates heavy computational load to construct a sufficiently large dataset for training. Furthermore, results obtained from a DNN do not have the theoretical support derived from the optimality conditions, which implies that local optimality is not guaranteed for the learning-based optimal control method. Reinforcement learning (RL) [31], combined with approximate dynamic programming, has been developed to search for optimal solutions with finite number of state and action pairs for both model based and model-free systems [32, 33]. However, the conventional RL methods are not well suited for optimal control problems with continuous states due to the dramatically increased computational load, difficulty of defining an appropriate reward function, and lack of convergence guarantee [34]. Although improved RL methods have been designed to handle the scalability issue of continuous action and state space, the data training process is still time and resource consuming with increased action dimensions [35, 36]. In [37], by combining RL with DNN, deep reinforcement learning is applied to learn a policy mapping the lander's estimated state directly to a commanded thrust for each engine, which can achieve good optimization performance for the six degree-of-freedom

power descent problem. However, though the existing learning based methods for solving FOPDG problems can achieve real-time optimization, heavy computational burden is required for offline training.

In this paper, a learning-based and theory-supported optimal control method (LB&TS OCM) is proposed. As opposed to a brute-force approach of applying SL directly to find the optimal solution, the learning process is guided by the necessary conditions derived from the Pontryagin's minimum principal. The combination of SL methods and fundamentals of optimal control theory dramatically reduces the dimension of the learning space by exploiting the features of the optimal solution. Based on Hamiltonian and first order necessary conditions, the optimal solution of the FOPDG problem is represented by a few critical parameters. Therefore, instead of learning all state and control variables, the proposed approach only need to learn the identified critical parameters. Once the critical parameters are determined, we are able to find the optimal control solution of FOPDG in continuous time and corresponding state variables by simply integrating the differential equations of vehicle dynamics forward to any desired time interval. Thus, the proposed LB&TS OCM aims to significantly reduce computational load required for dataset training when applied to solve the FOPDG problem.

This paper is organized as follows. In Section II, the formulation of the FOPDG problem is presented. Section III describes the proposed LB&TS OCM, including necessary conditions of optimality for FOPDG, data generation for training, and architecture of DNNs. In Section IV, simulation results are provided and compared with corresponding results obtained from the state-of-the-art method. Conclusions are addressed in Section V.

II. Problem Formulation

This section formulates the FOPDG problem. The following two assumptions are stated for the powered descent phase.

Assumption 1. *The rotation of Mars during the powered descent phase is ignored since the powered descent operation is completed within a short time interval compared with the length of the Mars day.*

Assumption 2. *The landing vehicle can be treated as a point mass.*

During the powered descent phase, the vehicle is required to decelerate and land on a preflight-designated surface location at near-zero speed with pinpoint precision. Motion during the powered descent phase is analyzed in a Cartesian coordinate system with the origin fixed to the surface of Mars. Without loss of generality, we assume the origin of the Cartesian coordinate is located at the landing point. The three dimensional (3D) equations of motion of the landing vehicle are expressed as

$$\dot{\mathbf{r}} = \mathbf{V} \tag{1a}$$

$$\dot{\mathbf{V}} = \mathbf{g} + \frac{\mathbf{T}_c}{m(t)} \tag{1b}$$

$$\dot{m} = -\alpha \|\mathbf{T}_c(t)\| \tag{1c}$$

where $\mathbf{r} = [x, y, z]^T$ and $\mathbf{V} = [v_x, v_y, v_z]^T$ denote the position and velocity vectors, respectively, α is a constant representing the combustion efficiency of a rocket, and $\mathbf{g}$ is the constant acceleration vector due to gravity. $\mathbf{T}_c = [\|\mathbf{T}_c(t)\| \sin\theta\cos\phi, \|\mathbf{T}_c(t)\| \sin\theta\sin\phi, \|\mathbf{T}_c(t)\| \cos\theta]$ is the thrust vector of the landing vehicle, where $\|\mathbf{T}_c(t)\|$ is the thrust magnitude, θ is the angle between the thrust vector and z-axis, and ϕ is the angle between the thrust projection on x-y plane and x-axis. Besides the nonlinear dynamics, there exist several control and state constraints, including the upper and lower bounded thrust magnitude, i.e. $T_{\min} \leq \|\mathbf{T}_c(t)\| \leq T_{\max}$, and the glide-slope constraint on the position vector, as shown in Fig. 1. The glide-slope constraint ensures that the vehicle is kept within a safe distance from the ground until it reaches the landing point. The glide-slope constraint can be described as a convex cone, expressed as

$$C = \{\mathbf{r} \in \mathbb{R}^2 : \frac{(\mathbf{r} - \mathbf{r}_f)^T \mathbf{c}}{\|\mathbf{r} - \mathbf{r}_f\| \cdot \|\mathbf{c}\|} \geq \cos(\beta)\}, \tag{2}$$

where $\mathbf{c}$ is the unit vector of glide-slope cone axis, β is the half-angle of the glide-slope cone, and $\mathbf{r}_f$ is the position vector of the landing point. Based on the assumption that the origin of the Cartesian coordinate is located at the landing point, we have $\mathbf{c} = [0, 0, 1]^T$ and $\mathbf{r}_f = [0, 0, 0]^T$. Then, the glide-slope constraint in (2) can be simplified as

$$z^2 \sin^2\beta - (x^2 + y^2)\cos^2\beta \geq 0. \tag{3}$$

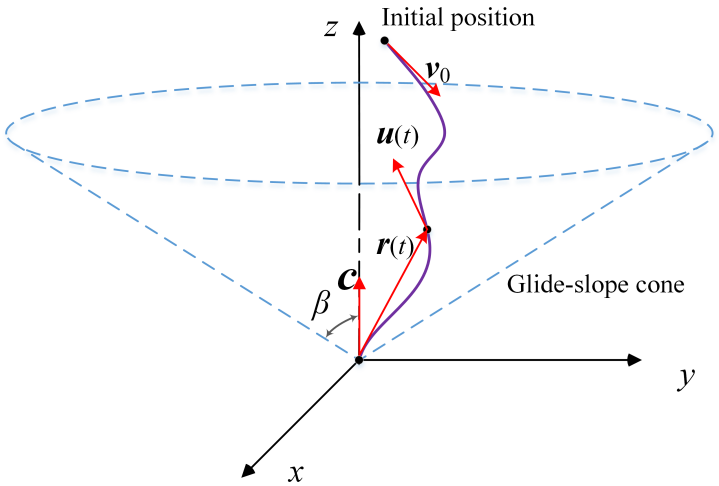


Fig. 1 Illustration of the glide-slope constraint

Therefore, the FOPDG problem can be formulated as

$$
\begin{aligned}
&\min_{t_f,\mathbf{T}_c} \int_{t_0}^{t_f} \alpha\|\mathbf{T}_c(t)\|dt, && \text{Objective}\\
\text{subject to}\quad &\dot{\mathbf{r}} = \mathbf{V},\ \dot{\mathbf{V}} = \mathbf{g} + \tfrac{\mathbf{T}_c}{m(t)},\ \dot{m} = -\alpha\|\mathbf{T}_c(t)\| && \text{Vehicle dynamic constraint}\\
&0 \le T_{\min} \le \|\mathbf{T}_c(t)\| \le T_{\max}, && \text{Bounded constraint on control magnitude}\\
&\mathbf{r}(t) \in C,\ \forall t \in [t_0, t_f], && \text{Glide-slope constraint}\\
&m(t_0) = m_{\text{wet}},\ m(t) \ge m_{\text{dry}}, && \text{Mass constraint}\\
&\mathbf{r}(t_0) = \mathbf{r}_0,\ \mathbf{V}(t_0) = \mathbf{V}_0, && \text{Initial conditions}\\
&\mathbf{r}(t_f) = \mathbf{0},\ \mathbf{V}(t_f) = \mathbf{0}. && \text{Terminal conditions}
\end{aligned}
\tag{4}
$$

where m_{wet} is the gross mass of the vehicle at the beginning of the powered descent phase, and m_{dry} is the structural mass. Our goal is to find the control vectors $\mathbf{T}_c(t)$ along the powered descent phase and its duration t_f to minimize the overall fuel consumption $\int_{t_0}^{t_f} \alpha\|\mathbf{T}_c(t)\|dt$ while satisfying all specified constraints. Extensive methods, especially NLP based methods, have been developed to solve the FOPDG problem with both numerical and experimental verification. We propose an alternative method that combines learning and the structure of the optimal solution to obtain a continuous-time optimal control solution in real-time.

III. Learning-based and Theory-Supported Optimal Control Method

A. Necessary Conditions of Optimality

We start with deriving the Hamiltonian and necessary conditions of optimality for the powered descent guidance problem expressed in (4). For the FOPDG problem formulated in (4), the augmented Hamiltonian is expressed as

$$
\begin{aligned}
H =&\lambda_1 v_x + \lambda_2 v_y + \lambda_3 v_z + \lambda_4 \frac{\|\mathbf{T}_c(t)\|\sin\theta\cos\phi}{m} + \lambda_5 \frac{\|\mathbf{T}_c(t)\|\sin\theta\sin\phi}{m} + \lambda_6(-g + \frac{\|\mathbf{T}_c(t)\|\cos\theta}{m})\\
&- \lambda_7\alpha\|\mathbf{T}_c(t)\| + \mu_1(z^2\sin^2\beta - (x^2+y^2)\cos^2\beta) + \mu_2(m - m_{dry})
\end{aligned}
\tag{5}
$$

and the first-order conditions of optimality are written as

$$
\frac{\partial H}{\partial\|\mathbf{T}_c(t)\|} = \lambda_4\frac{\sin\theta\cos\phi}{m} + \lambda_5\frac{\sin\theta\sin\phi}{m} + \lambda_6\frac{\cos\theta}{m} - \lambda_7\alpha \tag{6a}
$$

$$
\frac{\partial H}{\partial\theta} = \frac{\|\mathbf{T}_c(t)\|}{m}(\lambda_4\cos\theta\cos\phi + \lambda_5\cos\theta\sin\phi - \lambda_6\sin\theta) = 0 \tag{6b}
$$

$$
\frac{\partial H}{\partial\phi} = \frac{\|\mathbf{T}_c(t)\|}{m}\sin\theta(\lambda_5\cos\phi - \lambda_4\sin\phi) = 0 \tag{6c}
$$

where $\boldsymbol{\lambda} = [\lambda_1, \ldots, \lambda_7]^T$ is the adjoint variable set associated with the dynamic constraints and $\boldsymbol{\mu} = [\mu_1, \mu_2]^T$ is the Lagrange multiplier set associated with the state inequality constraints. The optimality condition in (6a) does not explicitly contain the thrust magnitude $\|\mathbf{T}_c(t)\|$, which yields the optimal control solution, expressed as

$$\|\mathbf{T}_c(t)\| = \begin{cases} T_{\min}, & \frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} < 0 \\ T_{\max}, & \frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} > 0 \\ \in [T_{\min}, T_{\max}], & \frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} = 0 \end{cases} \tag{7}$$

To show that the the optimal solution of the thrust magnitude is a bang-bang control, the following Lemma is proved by referring to the idea of [38].

Lemma 1. *For the FOPDG problem defined in* (4)*, the optimal solution of the thrust magnitude is a bang-bang control when the glide-slope constraint is inactive.*

Proof. According to equation (7), the optimal control magnitude may not be an exact bang-bang control. We assume that there exists $\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} = 0$ for a non-trivial period of time for the optimal solution. Then, during the period when $\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} = 0$, we have $\frac{d}{dt}(\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|}) = 0$. Let $\lambda_{\mathbf{r}} = [\lambda_1, \lambda_2, \lambda_3]^T$, $\lambda_{\mathbf{v}} = [\lambda_4, \lambda_5, \lambda_6]^T$, then (6a) can be rewritten as

$$\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} = \frac{{\lambda_{\mathbf{v}}}^T \mathbf{1}_T}{m} - \lambda_7 \alpha \tag{8}$$

where $\mathbf{1}_T$ is the unit control vector maximizing H, and it is determined by $\mathbf{1}_T = \frac{\lambda_{\mathbf{v}}}{\|\lambda_{\mathbf{v}}\|}$ [22]. By replacing $\mathbf{1}_T$, (8) can be reformulated as

$$\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} = \frac{\|\lambda_{\mathbf{v}}\|}{m} - \lambda_7 \alpha \tag{9}$$

Then $\frac{d}{dt}(\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|}) = 0$ can be expressed as

$$\frac{d}{dt}(\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|}) = -\frac{{\lambda_{\mathbf{v}}}^T \lambda_{\mathbf{r}}}{m\|\lambda_{\mathbf{v}}\|} = 0 \tag{10}$$

Because the glide-slope constraint is inactive by assumption, from the first-order necessary conditions of optimality, we have $\dot{\lambda_{\mathbf{r}}} = -\frac{\partial H}{\partial \mathbf{r}} = \mathbf{0}$, $\dot{\lambda_{\mathbf{v}}} = -\frac{\partial H}{\partial \mathbf{V}} = -\lambda_{\mathbf{r}}$, $\dot{\lambda}_7 = -\frac{\partial H}{\partial m} = \frac{{\lambda_{\mathbf{v}}}^T \mathbf{T}_c(t)}{m^2}$ [38]. Therefore, $\lambda_{\mathbf{r}}$ is a constant vector and $\lambda_{\mathbf{v}} = \lambda_{\mathbf{r}} t + \mathbf{c_a}$, where $\mathbf{c_a}$ is an unknown constant vector. Then $\frac{d}{dt}(\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|}) = 0$ can be expressed as

$$\frac{d}{dt}(\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|}) = -\frac{{\lambda_{\mathbf{v}}}^T \lambda_{\mathbf{r}}}{m\|\lambda_{\mathbf{v}}\|} = \frac{{\lambda_{\mathbf{r}}}^2 t - {\lambda_{\mathbf{r}}}^T \mathbf{c_a}}{m\|\lambda_{\mathbf{v}}\|} = 0 \tag{11}$$

To let (11) hold, we have $\lambda_{\mathbf{r}} = \mathbf{0}$ and $\mathbf{c_a} \neq 0$. For the free-time problem, the transversality condition for optimality is $H(t_f) = 0$. By replacing $\lambda_{\mathbf{r}} = \mathbf{0}$ and $\mathbf{c_a} \neq 0$ into $H(t_f)$, it leads to $H(t_f) = \mathbf{c_a}^T \mathbf{g} \neq 0$, which contradicts the transversality condition $H(t_f) = 0$. Therefore, when the glide-slope constraint is inactive, the relation $\frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} = 0$ cannot hold for a non-trivial period of time for the optimal solution, which means the optimal solution of the thrust magnitude is a bang-bang control. □

Therefore, according to equation (7) and Lemma 1, the optimal solution of the thrust magnitude is a bang-bang control when the the glide-slope constraint is inactive and the singular arc will not exist, which can be expressed as

$$\|\mathbf{T}_c(t)\| = \begin{cases} T_{\min}, & \frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} < 0 \\ T_{\max}, & \frac{\partial H}{\partial \|\mathbf{T}_c(t)\|} > 0 \end{cases} \tag{12}$$

When the glide-slope constraint (3) is active, the position of vehicle can be expressed as $[Z\cos\gamma\tan\beta, Z\sin\gamma\tan\beta, Z]$, where Z is the distance from the x-y plane to the vehicle, γ is the angle between the x axis and the x-y projection of vector $(\mathbf{r} - \mathbf{r}_f)$. Then we can obtain the following proposition:

Proposition 1. *For an optimal solution of the 3D FOPDG problem in* (4)*, the glide-slope constraint will only be active with nontrivial duration* $\Delta t > 0$ *when the following equation is satisfied:*

$$\frac{1}{\sqrt{(\tan^2\beta(\ddot{Z}^2 + 4\dot{Z}^2\dot{\gamma}^2 + Z^2\dot{\gamma}^4 + Z^2\ddot{\gamma}^2 - 2Z\ddot{Z}\dot{\gamma}^2 + 4\dot{Z}Z\ddot{\gamma}\dot{\gamma}) + \ddot{Z}^2 + 2g\ddot{Z} + g^2)}} = \frac{m}{\|\mathbf{T}_c(t)\|}. \tag{13}$$

Proof. We assume that the vehicle remains on the glide-slope for time duration $\Delta t > 0$. Then, the velocity components of the vehicle are expressed as

$$v_{Gx} = \frac{\partial(Z\cos\gamma\tan\beta)}{\partial t} = (\dot{Z}\cos\gamma - Z\sin\gamma\dot{\gamma})\tan\beta, \tag{14}$$

$$v_{Gy} = \frac{\partial(Z\sin\gamma\tan\beta)}{\partial t} = (\dot{Z}\sin\gamma + Z\cos\gamma\dot{\gamma})\tan\beta, \tag{15}$$

$$v_{Gz} = \frac{\partial Z}{\partial t} = \dot{Z}, \tag{16}$$

where $v_{Gx},\ v_{Gy},\ v_{Gz}$ are the vehicle components in $x,\ y,\ z$ directions, respectively. From the time derivatives of $v_{Gx},\ v_{Gy},\ v_{Gz}$, we have the acceleration of the vehicle along $x,\ y,\ z$ axis expressed as

$$a_{Gx} = \frac{\partial v_{Gx}}{\partial t} = (\ddot{Z}\cos\gamma - 2\dot{Z}\sin\gamma\dot{\gamma} - Z\cos\gamma\dot{\gamma}^2 - Z\sin\gamma\ddot{\gamma})\tan\beta, \tag{17}$$

$$a_{Gy} = \frac{\partial v_{Gy}}{\partial t} = (\ddot{Z}\sin\gamma + 2\dot{Z}\cos\gamma\dot{\gamma} - Z\sin\gamma\dot{\gamma}^2 + Z\cos\gamma\ddot{\gamma})\tan\beta, \tag{18}$$

$$a_{Gz} = \frac{\partial v_{Gz}}{\partial t} = \ddot{Z}. \tag{19}$$

As the thrust vector of the landing vehicle is $\mathbf{T}_c = [\|\mathbf{T}_c(t)\| \sin\theta\cos\phi, \|\mathbf{T}_c(t)\| \sin\theta\sin\phi, \|\mathbf{T}_c(t)\| \cos\theta]$, then we have $a_{Gx} = \frac{\|\mathbf{T}_c(t)\| \sin\theta\cos\phi}{m}$, $a_{Gy} = \frac{\|\mathbf{T}_c(t)\| \sin\theta\sin\phi}{m}$, $a_{Gz} = \frac{\|\mathbf{T}_c(t)\| \cos\theta}{m} - g$, which can be expressed as

$$\begin{aligned} \|\mathbf{T}_c(t)\|^2 &= ((a_{Gx})^2 + (a_{Gy})^2 + (a_{Gz} + g)^2)m^2 \\ &= (\tan^2\beta(\ddot{Z}^2 + 4\dot{Z}^2\dot{\gamma}^2 + Z^2\dot{\gamma}^4 + Z^2\ddot{\gamma}^2 - 2Z\ddot{Z}\dot{\gamma}^2 + 4\dot{Z}Z\ddot{\gamma}\dot{\gamma}) + \ddot{Z}^2 + 2g\ddot{Z} + g^2)m^2, \end{aligned} \tag{20}$$

which is equivalent to

$$\frac{1}{\sqrt{(\tan^2\beta(\ddot{Z}^2 + 4\dot{Z}^2\dot{\gamma}^2 + Z^2\dot{\gamma}^4 + Z^2\ddot{\gamma}^2 - 2Z\ddot{Z}\dot{\gamma}^2 + 4\dot{Z}Z\ddot{\gamma}\dot{\gamma}) + \ddot{Z}^2 + 2g\ddot{Z} + g^2)}} = \frac{m}{\|\mathbf{T}_c(t)\|}. \tag{21}$$

Therefore, we conclude that the glide-slope constraint is active with $\Delta t > 0$ if equation (21) is satisfied. This completes proof of Proposition 1. □

Moreover, when the glide-slope constraint is active, the optimal $\|\mathbf{T}_c(t)\|$ can either be singular or bang-bang control. When $\|\mathbf{T}_c(t)\|$ is singular, it satisfies $\frac{d^K}{dt^K}(\frac{\partial H}{\partial\|\mathbf{T}_c(t)\|}) = 0$, where $K \geq 0$ refers to the minimum index when $\frac{d^K}{dt^K}(\frac{\partial H}{\partial\|\mathbf{T}_c(t)\|})$ includes an explicit $\|\mathbf{T}_c(t)\|$. For the studied FOPDG problem, we have $K = 4$ and

$$\frac{d^4}{dt^4}(\frac{\partial H}{\partial\|\mathbf{T}_c(t)\|}) = \ddot{\lambda}_{\mathbf{r}}\mathbf{1}_T - 2\frac{\ddot{\lambda}_{\mathbf{r}}\lambda_{\mathbf{r}}}{\|\lambda_{\mathbf{v}}\|} - \frac{\|\dot{\lambda}_{\mathbf{r}}\|^2}{\|\lambda_{\mathbf{v}}\|} + \frac{d^2}{dt^2}(\frac{\|\lambda_{\mathbf{r}}\|^2}{\|\lambda_{\mathbf{v}}\|}) = 0, \tag{22}$$

where $\ddot{\lambda}_{\mathbf{r}} = 2\mathbf{M_1}\cos^2\beta(\mu_1(\|\mathbf{T}_c(t)\|\mathbf{1}_T - \mathbf{g})/m + 2\dot{\mu}_1\mathbf{V} + \ddot{\mu}_1\mathbf{r})$ is the only term in (22) which includes an explicit $\|\mathbf{T}_c(t)\|$, and $\mathbf{M_1} \in \mathbb{R}^{3\times3}$ is a constant coefficient matrix. Then we have

$$\|\mathbf{T}_c(t)\| = \frac{m\mathbf{f} + \mu_1\mathbf{g}}{\mu_1\mathbf{1}_T}, \tag{23}$$

where $\mathbf{f} = (2\frac{\ddot{\lambda}_{\mathbf{r}}\lambda_{\mathbf{r}}}{\|\lambda_{\mathbf{v}}\|} + \frac{\|\dot{\lambda}_{\mathbf{r}}\|^2}{\|\lambda_{\mathbf{v}}\|} - \frac{d^2}{dt^2}(\frac{\|\lambda_{\mathbf{r}}\|^2}{\|\lambda_{\mathbf{v}}\|}))/(2\mathbf{M_1}\cos^2\beta\mathbf{1}_T) - 2\dot{\mu}_1\mathbf{V} - \ddot{\mu}_1\mathbf{r}$. Considering the radical expression of (21), we can conclude that equation (21) is not easily to hold while satisfying the optimality conditions when $\|\mathbf{T}_c(t)\|$ is singular.

On the other hand, if the optimal $\|\mathbf{T}_c(t)\|$ is bang-bang control when the glide-slope constraint is active, then $\|\mathbf{T}_c(t)\|$ in (21) will be constant for a time interval. In addition, $\dot{m}$ in (1) is a constant, which means m is a linear function with respect to time. Meanwhile, considering the high non-linearity of the left side of (21), we can conclude that equation (21) is not easily to hold while satisfying the optimality conditions. For the studied FOPDG problem, we found that the equation (21) does not hold for any cases in the range specified in Section III.B.1. Therefore, the glide-slope constraint will only be active at isolated points with duration $\Delta t \to 0$ in the studies cases.

Remark 1. *Proposition 1 and the above analysis indicate that the glide-slope constraint is only active at instances in time for general cases. Combined with Lemma 1, we claim that the optimal solution of the thrust magnitude is a*

bang-bang control and the glide slope will only be active with duration $\Delta t \to 0$.

For a inequality constraint, if its qth time derivative is an explicit function of control variables and at the same time its ith ($0 \leq i < q$, i is a natural number) time derivative is not an explicit function of control variables, the state inequality constraint is called qth-order state variable inequality constraint. Work in [39] claims that when the optimal problem only has first-order state variable inequality constraints, its adjoint variables will be continuous at the isolated point. However, for the optimal control problem with second-order inequality constraints, the adjoint variables are not necessarily continuous at the isolated point [40]. Considering that the glide-slope constraint belongs to the second-order inequality constraint, we can conclude that the adjoint variables λ_i, $i = 1, \ldots, 6$ are not necessarily continuous at the isolated points. Therefore, λ_i, $i = 1, \ldots, 6$, are handled as piecewise functions and the switching points are the isolated points.

For the optimal control, when an inequality constraint is inactive, the corresponding Lagrange multiplier equals to zero. Thus, for the FOPDG problem, we have $\mu_1 = 0$ when the glide-slope constraint (2) is inactive. Therefore, when the glide-slope constraint is inactive, combining with the necessary conditions of adjoint variables, we have $\dot{\boldsymbol{\lambda}} = -\frac{\partial H}{\partial \mathbf{r}}$, which can be extended as

$$\dot{\lambda}_1 = -\frac{\partial H}{\partial x} = 0 \tag{24a}$$

$$\dot{\lambda}_2 = -\frac{\partial H}{\partial y} = 0 \tag{24b}$$

$$\dot{\lambda}_3 = -\frac{\partial H}{\partial z} = 0 \tag{24c}$$

$$\dot{\lambda}_4 = -\frac{\partial H}{\partial v_x} = -\lambda_1 \tag{24d}$$

$$\dot{\lambda}_5 = -\frac{\partial H}{\partial v_y} = -\lambda_2 \tag{24e}$$

$$\dot{\lambda}_6 = -\frac{\partial H}{\partial v_z} = -\lambda_3 \tag{24f}$$

By integrating (24), it is straight forward to find the solutions of adjoint variables, written as

$$\lambda_1 = C_1,\ \lambda_2 = C_2,\ \lambda_3 = C_3,\ \lambda_4 = -C_1 t + C_4,\ \lambda_5 = -C_2 t + C_5,\ \lambda_6 = -C_3 t + C_6, \tag{25}$$

where C_i, $i = 1, \ldots, 6$, are constant and can be determined from the value of $\boldsymbol{\lambda}$ at the initial point. Combining Proposition 1 with Lemma 1, we can conclude that the adjoint variables λ_i, $i = 1, \ldots, 6$, are piecewise functions and the piecewise points are the isolated points. Then, for each segment between the isolated point, via combining (25), (6b)

and (6c), we have

$$\tan\phi = \frac{\lambda_5}{\lambda_4} \tag{26a}$$

$$\tan\theta = \frac{\lambda_4}{\lambda_6}(\cos\phi + \frac{\lambda_5}{\lambda_4}\sin\phi) = \frac{\lambda_4}{\lambda_6}(\cos\phi + \tan^2\phi\cos\phi) = \frac{\lambda_4}{\lambda_6\cos\phi}. \tag{26b}$$

By substituting the value of $\boldsymbol{\lambda}$ in (26) using the solution obtained from (25), the optimal thrust direction angles, ϕ and θ can be determined by

$$\tan\phi = \frac{-C_2t + C_5}{-C_1t + C_4} \tag{27a}$$

$$\cos\phi\tan\theta = \frac{-C_1t + C_4}{-C_3t + C_6}. \tag{27b}$$

Therefore, when C_i, $i = 1, \ldots, 6$, are determined, we can compute ϕ and θ accordingly.

Besides, with $\mu_1 = 0$, the optimality conditions of the FOPDG problem with the glide-slope constraint is the same as the FOPDG problem without the glide-slope constraint. In literature [22, 41], it has been rigorously proved that under the constant-gravity assumption, the optimal solution of the minimum-fuel powered descent guidance problem without the glide-slope constraint has at most three sub-arcs, with the thrust magnitude in the order of max-min-max. In other words, there are at most two switching time points, shown in Fig. 2, where t_{s1} denotes the switching time from maximum thrust magnitude to minimum thrust magnitude, t_{s2} is the switching time from minimum thrust magnitude to maximum thrust magnitude, and t_f is the terminal time. For the cases that have only one sub-arc, t_{s1} and t_{s2} are set as the terminal time t_f if it is the maximum sub-arc, otherwise t_{s1} is set as the starting time t_0 and t_{s2} is set as the terminal time t_f. For cases that have two sub-arcs, t_{s2} is set as the terminal time t_f if the two sub-arc are in the order of max-min, otherwise t_{s1} is set as the starting time t_0. Therefore, once t_{s1}, t_{s2} and t_f are obtained, the optimal control profile of the thrust magnitude, $\|\mathbf{T}_c(t)\|$, during the powered descent phase can be determined.

From the above analysis on necessary conditions, the optimal guidance law for the 3D FOPDG problem can be determined once the following parameters, named critical parameters, are obtained:

1) The switching time t_{s1} and t_{s2} for the bang-bang control;
2) Duration of the powered descent phase t_f;
3) The glide-slope boundary touching time t_b for each segment;
4) The coefficients $C_1, C_2, C_3, C_4, C_5, C_6$ for each segment.

Note that, if the glide-slope constraint is not active along the optimized trajectory, the fitted curves for $\tan\phi^*$ and $\tan\theta^*\cos\phi^*$ only contain one segment, and the glide-slope boundary touching time t_b is set as t_f. After determining the above identified critical parameters, we can find the control switching time points from t_{s1}, t_{s2} and the duration of descent phase from t_f. Meanwhile, the piecewise thrust direction angles θ and ϕ can be determined from corresponding

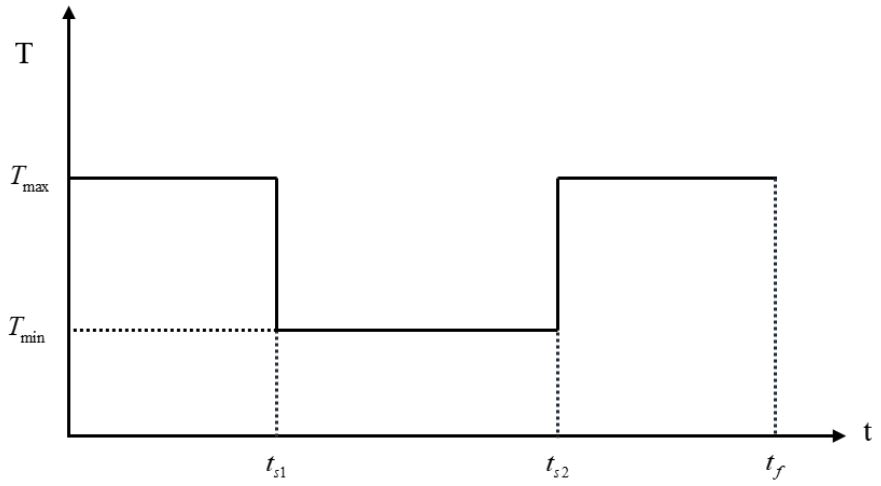


Fig. 2 Bang-bang control and switching time t_{s1}, t_{s2}

$C_1, C_2, C_3, C_4, C_5, C_6$ according to (27). Then the time-continuous optimal control vector $\mathbf{T}_c(t)$ during the powered descent phase can be determined according to the switching time points and thrust direction angles θ and ϕ. With the optimal control profile determined, the state variables in continuous time can be obtained by integrating the vehicle dynamics in (1) from t_0 to t_f.

B. Critical Parameter Determination from Deep Learning

1. Generation of Training Dataset

Based on the necessary conditions of optimality derived for the FOPDG problem, the DNNs are applied to map the functions between problem inputs and identified critical parameters. In other words, given a set of problem inputs, we aim to find the optimal solution of FOPDG from the trained neural networks. Instead of finding all state and control variables of the optimal solution, the proposed LB&TS OCM only needs to find the identified critical parameters. As described above, once these critical parameters are determined, the complete optimal solution, including control and state variables in continuous time, can be obtained. For the 3D FOPDG problem, the inputs of the training dataset include $\mathbf{X}_{\text{input}} = [x(t_0), v_x(t_0), y(t_0), v_y(t_0)]$ that represent the state variables at the initial point.

To generate an offline training database, an NLP solver is applied to solve the FOPDG problem. The commercial Tomlab-SNOPT solver which combines collocation and interior point method is elected here to solve the FOPDG problem[42]. This NLP solver is well-known due to its easy implementation interface and the pseudospectral collocation scheme that adjusts the location of discrete nodes to improve the solution precision. The training dataset is generated offline and the generation procedure includes two steps: (a) Generating optimal solutions with discrete control and state variables from the NLP; and (b) calculating the corresponding critical parameters for every validated NLP solution. For a given input set $\mathbf{X}_{\text{input}}$ that specifies the initial conditions, the control and state variables of the FOPDG problem are

discretized into N discrete nodes and then solved by an NLP solver. From the NLP solution, the critical parameters t_{s1}, t_{s2}, t_f and the optimal control variables $\|\mathbf{T}_c\|^*$, θ^* and ϕ^* can be determined directly. Then, by calculating the isolated points, the glide-slope touching time t_b for each segment can be obtained. Moreover, with the obtained θ^* and ϕ^*, $\tan\phi^*$ and $\cos\phi^*\tan\theta^*$ associated with each discrete node can be calculated. Finally, via combining the obtained optimal solutions and (27), the remaining critical parameters can be determined by curve fitting $\tan\phi^*$ and $\cos\phi^*\tan\theta^*$ obtained at all discrete nodes. To further improved the accuracy of the pinpoint landing, a series of optimization problems are formulated to determine $C_1, C_2, C_3, C_4, C_5, C_6$ for each segment. First, we need to divide each segment equally into N_k intervals. For instance, for the first segment between the starting point and the first isolated point, the duration is $t_b - t_0$ and the discrete time step is $(t_b - t_0)/(N_k)$. Then, to satisfy the glide-slope constraint and terminal landing conditions, the errors between the glide-slope boundary and points on the optimal trajectory and terminal errors are considered, which can be expressed as

$$E_{rr} = \sum_{i=1}^{N_k+1} \max(0, x(i)^2 + y(i)^2 - z(i)^2\tan^2\beta) + \|\mathbf{V}(N_k+1) - \mathbf{V}_f^j\|^2 + \|\mathbf{r}(N_k+1) - \mathbf{r}_f^j\|^2 \tag{28}$$

By integrating the terminal landing error and the glide-slope constraint, the curve fitting problem can be formulated as an optimization problem. With the terminal conditions and glide-slope constraint considered in the objective function of curve fitting, a set of $C_1, C_2, C_3, C_4, C_5, C_6$ for each segment can be obtained with precise landing and the glide-slope constraint satisfied.

In this paper, the range of inputs is set as $x(t_0) \in [-0.5, 2]$ km, $v_x(t_0) \in [-100, 100]$ m/s, $y(t_0) \in [-0.5, 0.5]$ km, $v_y(t_0) \in [-75, 75]$ m/s. While dealing with the collected solutions, we found that for the studied cases, there are at most one isolated point for all the solved cases. The identified critical parameters are then reduced. All generated cases in the database are classified into two categories. If the glide-slope constraint is not active along the optimized trajectory, the fitted curves for $\tan\phi^*$ and $\tan\theta^*\cos\phi^*$ only contain one segment, and the glide-slope boundary touching time t_b is set as t_f. On the other hand, if the glide-slope constraint is active at one isolated point, we use two segments to fit $\tan\phi^*$ and $\tan\theta^*\cos\phi^*$. One starts from the initial point to the first isolated point and the other one starts from the first isolated point to the landing point, as shown in Fig. 3. There critical parameters are summarized below.

1) The switching time t_{s1} and t_{s2} for the bang-bang control;
2) Duration of the powered descent phase t_f;
3) The glide-slope boundary touching time t_b;
4) The coefficients $C_1^1, C_2^1, C_3^1, C_4^1, C_5^1, C_6^1, C_1^2, C_2^2, C_3^2, C_4^2, C_5^2, C_6^2$.

Then we denote $\mathbf{X}_{\text{output}} = [C_1^1, C_2^1, C_3^1, C_4^1, C_5^1, C_6^1, C_1^2, C_2^2, C_3^2, C_4^2, C_5^2, C_6^2, t_{s1}, t_{s2}, t_b, t_f]$. Using the above identified parameters, the control thrust $\|\mathbf{T}_c(t)\|$ can be determined from t_{s1}, t_{s2}, and the duration of descent phase t_f. Meanwhile, the thrust direction angles θ and ϕ before and after the isolated point can be determined from t_b and

$C_1^1, C_2^1, C_3^1, C_4^1, C_5^1, C_6^1, C_1^2, C_2^2, C_3^2, C_4^2, C_5^2, C_6^2$ according to (27). Next, similar to the description in the last paragraph of Section III.A, the complete solution of the FOPDG problem, including time-continuous control and state variables can be obtained.

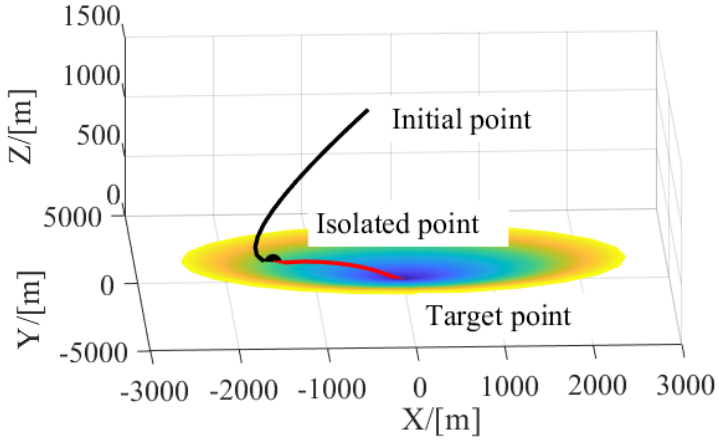
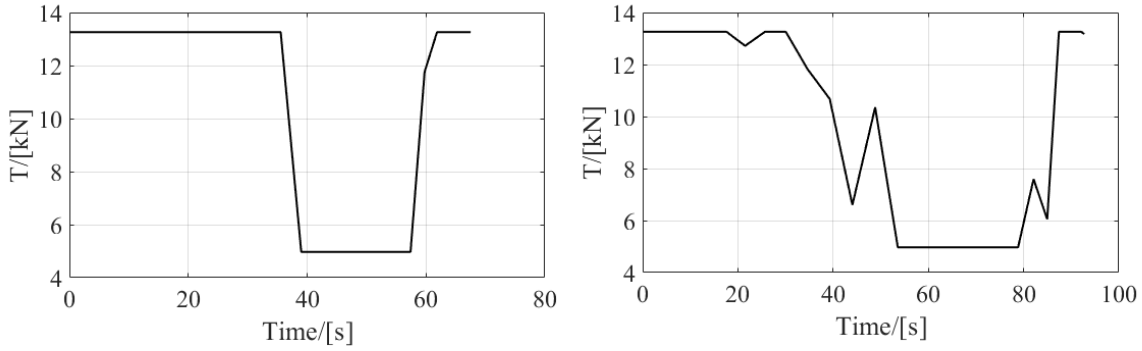


Fig. 3 Illustration of two piece-wise continuous segments and isolated point of a powered descent trajectory: the black solid line is the segment before the isolated point, and the red solid line is the segment after the isolated point

When using NLP to find the optimal solution of the FOPDG problem with a given input set $\mathbf{X}_{\text{input}}$, not all solutions from NLP can be utilized in the dataset for training. To explain this issue, two examples of NLP solutions are illustrated in Fig. 4, where the solution in Fig. 4a shows a thrust profile close to Bang-Bang control while the solution in Fig. 4b is not feasible to be implemented although the NLP solver indicates that the solution converges. One reason that leads to this phenomena is the inappropriate selection of discretion number and nodes. Therefore, after excluding non-convergent solutions from NLP, solutions that do not yield an approximate Bang-Bang control are also excluded from the dataset for training to improve the accuracy of learning from DNNs.



(a) Bang-bang Solution from NLP **(b) NLP solution not suitable for learning**

Fig. 4 Examples of FOPDG solutions from NLP

2. Architecture of DNNs

After generating a database with a sufficient number of cases from NLP, the DNNs can be constructed. The architecture of a constructed DNN for the 3D FOPDG problem is shown in Fig. 5, where L fully connected layers are included and each layer has m_l neurons except the output layer. The number of neurons in the output layer is the same as the length of $\mathbf{X}_{\text{output}}$. To better fit the nonlinearity of different networks, the activation function $\mathbf{F}_{\text{act}}$ of each layer will be adjusted independently.

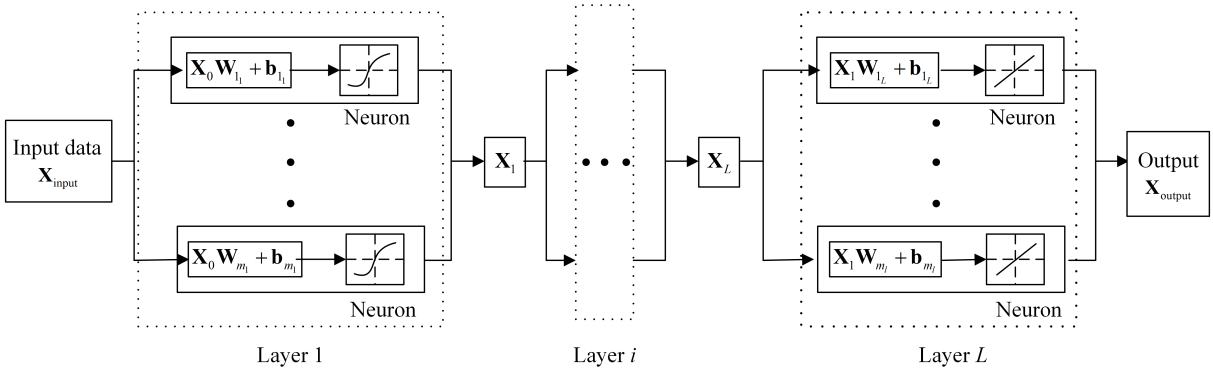
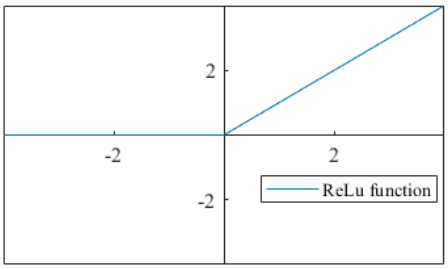


Fig. 5 The architecture of a constructed DNN

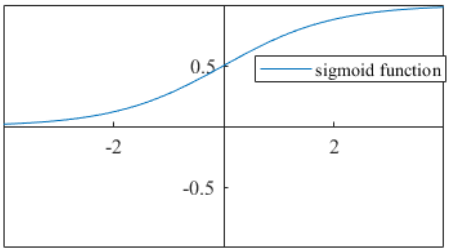
Typically, there exists three commonly used activation functions, including rectified linear unit (ReLu) function, hyperbolic isolated function, and sigmoid function, illustrated in Fig. 6. The sigmoid function is positive but not zero-centered, whereas the hyperbolic isolated function transfers inputs from interval $[-\infty, \infty]$ into the a bounded interval $[-1, 1]$. Meanwhile, the hyperbolic isolated function is sensitive to small changes in the middle part, which makes it a good candidate for training and calculating a function gradient. However, insensitivity of gradient exists close to the edges of the hyperbolic isolated function. For ReLu function, due to its simple formulation, it has faster convergence rate than the other two types of activation functions. By combining these commonly used activation functions, the DNNs can be constructed to map the relationship between $\mathbf{X}_{\text{input}}$ and $\mathbf{X}_{\text{output}}$.

3. Training results of DNNs

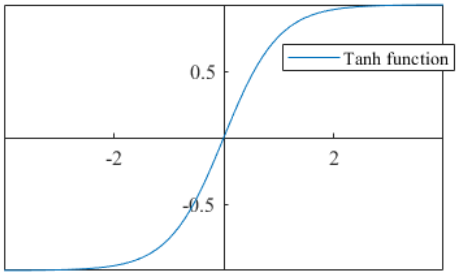
To improve the learning accuracy, sixteen independent DNNs are constructed to learn the sixteen identified critical parameters independently. In other words, with the same network architecture, each DNN will produce one of the sixteen identified critical parameters as output. Every DNN has four hidden layers and the number of neurons in each layer are set as 10, 20, 20, and 10, respectively. In addition, the corresponding activation functions include three hyperbolic tangent functions and one ReLu. The range of inputs is set as $x(t_0) \in [-0.5, 2]$ km, $v_x(t_0) \in [-100, 100]$ m/s, $y(t_0) \in [-0.5, 0.5]$ km, $v_y(t_0) \in [-75, 75]$ m/s. In addition, 9317 cases are generated offline and solved by the NLP solver, where only 6748 cases obtain a convergent solution. Moreover, among the convergent cases, solutions that not suitable for learning, such as the case in Fig. 4b, are excluded. As a result, the dataset size for the 3D FOPDG problem



(a) ReLu function



(b) Sigmoid function



(c) Hyperbolic tangent function

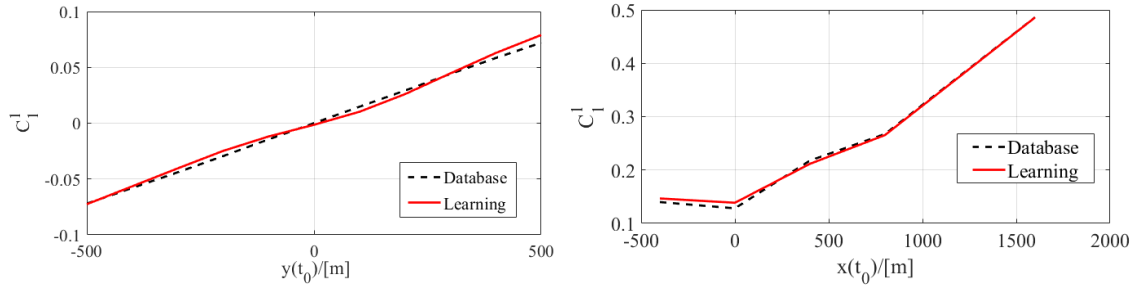
Fig. 6 Activation functions in neural networks

is 4410, where 15% of them are handled as validating dataset, 10% of them are handled as testing dataset, and the other 75% of them are handled as training dataset. Table 1 lists the number of epochs at convergence and the mean square error (MSE) of all DNNs for training the sixteen identified critical parameters. Here, an epoch is a full pass through the entire dataset. In addition, the learning results of t_f and C_1 are illustrated in Figs. 7 and 8, where the red solid line represents the solutions from a well trained DNN, and the black dash line represents the refined solution from the NLP solver. Since the validating data, testing data, and training data are randomly distributed from the full dataset, all these data sets are included when the full range of inputs $y(t_0)$, $x(t_0)$, $v_x(t_0)$ and $v_y(t_0)$ are considered in Figs. 7 and 8. In Table 1, the MSE of all critical parameters are relatively small. The only MSE item larger than $1.0e^{-3}$ is the one of t_{s1}. Moreover, the number of epochs at convergence for all sixteen DNNs are less than 3000.

Compared with the learning-based optimal control method in [29] that focuses on learning the state-action pair for a given input set, the method proposed in this paper can achieve the same level training accuracy with a much smaller size of dataset. Specifically, work in [29] uses more than 10 million state-action pairs to train the neural network for a two-dimensional powered descent guidance problem with a narrower range of problem inputs, while only 4410 trajectories containing 4410 sets of critical parameters are generated as the training dataset using the proposed method. As a result, the proposed LB&TS OCM takes much less time for training with reduced memory, which shows obvious advantage for cases with wider distribution of problem inputs. Figure 7 illustrates the relationship between each initial state $x(t_0)$, $y(t_0)$, $v_x(t_0)$, $v_y(t_0)$ and the critical parameter C_1, restrictively. For instance, in figure 7a, it presents the

Table 1 Training results of DNNs for all critical parameters

Parameter	MSE of training dataset	MSE of validation dataset	MSE of testing dataset	Epochs
t_{s1}	$1.90e^{-3}$	$2.29e^{-3}$	$2.20e^{-3}$	1194
t_{s2}	$8.20e^{-4}$	$7.62e^{-4}$	$9.90e^{-4}$	1520
t_b	$9.42e^{-4}$	$8.68e^{-4}$	$9.37e^{-4}$	1920
t_f	$5.81e^{-4}$	$5.85e^{-4}$	$5.98e^{-4}$	2068
C_1^1	$7.86e^{-5}$	$8.02e^{-5}$	$9.64e^{-5}$	1821
C_2^1	$1.68e^{-4}$	$2.31e^{-4}$	$1.92e^{-4}$	2862
C_3^1	$2.13e^{-4}$	$2.84e^{-4}$	$2.86e^{-4}$	2158
C_4^1	$4.06e^{-5}$	$4.17e^{-5}$	$3.86e^{-5}$	1706
C_5^1	$8.77e^{-5}$	$9.73e^{-5}$	$1.04e^{-4}$	1928
C_6^1	$2.84e^{-4}$	$3.11e^{-4}$	$2.68e^{-4}$	2341
C_1^2	$3.16e^{-4}$	$3.69e^{-4}$	$3.21e^{-4}$	2750
C_2^2	$1.32e^{-4}$	$1.49e^{-4}$	$1.83e^{-4}$	1964
C_3^2	$5.68e^{-4}$	$5.90e^{-4}$	$7.82e^{-4}$	2410
C_4^2	$6.21e^{-4}$	$7.22e^{-4}$	$6.91e^{-4}$	1896
C_5^2	$2.35e^{-4}$	$2.66e^{-4}$	$3.94e^{-4}$	2050
C_6^2	$7.66e^{-4}$	$9.82e^{-4}$	$9.42e^{-4}$	1966



(a) The relationship between $y(t_0)$ and C_1^1

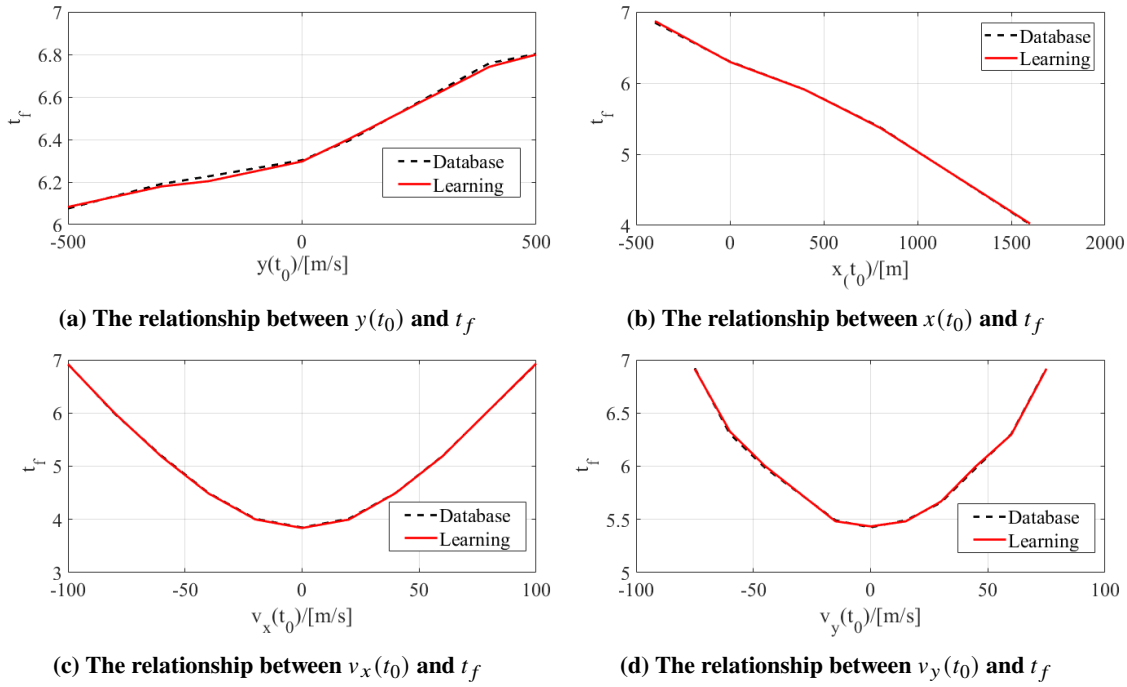
(b) The relationship between $x(t_0)$ and C_1^1

(c) The relationship between $v_x(t_0)$ and C_1^1

(d) The relationship between $v_y(t_0)$ and C_1^1

Fig. 7 Comparison of critical parameter C_1^1 between the database and learning results

relationship between $y(t_0) \in [-0.5, 0.5]$ km and C_1^1, where $x(t_0) = 0$ km, $v_x(t_0) = -100$ m/s, $v_y(t_0) = 60$ m/s are given. The red solid line and black dash line are close to each other, which further validates the accuracy of the learning



(a) The relationship between $y(t_0)$ and t_f

(b) The relationship between $x(t_0)$ and t_f

(c) The relationship between $v_x(t_0)$ and t_f

(d) The relationship between $v_y(t_0)$ and t_f

Fig. 8 Comparison of critical parameter t_f between the generated database and learning results

results and shows that the approximated functions between the inputs and outputs precisely represent the input-output relationship. Similarly, Fig. 8 validates the learning results of t_f under varying inputs. Additionally, results from Figs. 7 and 8 indicate that the approximated functions have simple formulations and do not have many inflection points, which further supports the statement that the scale of dataset required for the proposed LB&TS OCM can be significantly reduced. Further verification of efficiency and effectiveness of the LB&TS OCM is detailed in the following section.

IV. Simulation Results

To demonstrate the efficiency and effectiveness of the proposed LB&TS OCM, ten simulation cases of the FOPDG problem with different initial conditions are provided and compared to the corresponding solutions from NLP. Table 2 presents the initial and final conditions specified for the ten cases, where the first five cases are randomly selected from testing dataset and the other five are from the cases that cannot find a convergent solution using the NLP solver. Besides, the other problem parameters are specified as [38]

$$g = -3.711400 m/s^2,\ m_{\text{dry}} = 1505kg,\ m_{\text{wet}} = 1905kg, \tag{29}$$
$$T_{max} = 13.26kN,\ T_{min} = 4.972kN,\ \alpha = 4.35 \times 10^{-4}.$$

From the proposed method, we find the sixteen critical parameters from the constructed DNNs and then obtain the complete optimal solutions in continuous time.

Table 2 Initial and final conditions of five verification cases for powered descent guidance

	Initial conditions			Final conditions		
	$[x, y, z]$ km	$[v_x, v_y, v_z]$ m/s	m kg	$[x, y, z]$ km	$[v_x, v_y, v_z]$ m/s	m kg
$Case1$	$[2, -0.4, 1.5]$	$[-60, -75, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505
$Case2$	$[1.6, 0.2, 1.5]$	$[-100, 75, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505
$Case3$	$[1.2, 0.2, 1.5]$	$[20, 75, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505
$Case4$	$[0, -0.1, 1.5]$	$[0, -45, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505
$Case5$	$[-0.4, -0.5, 1.5]$	$[-100, 60, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505
$Case6$	$[-0.4, -0.1, 1.5]$	$[-80, -60, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505
$Case7$	$[-0.4, -0.1, 1.5]$	$[-80, 45, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505
$Case8$	$[0, 0.3, 1.5]$	$[-20, 75, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505
$Case9$	$[0.4, -0.1, 1.5]$	$[-100, 60, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505
$Case10$	$[0.8, 0.2, 1.5]$	$[40, 60, -75]$	1905	$[0, 0, 0]$	$[0, 0, 0]$	≥ 1505

When applying the proposed LB&TS OCM to solve these ten cases, each case can obtain the sixteen critical parameters from the constructed DNNs given the problem inputs. Then, the continuous-time state and control solutions can be determined accordingly. To demonstrate the performance of LB&TS OCM, the computational time and terminal states $[x(t_f), y(t_f), z(t_f), v_x(t_f), v_y(t_f), v_z(t_f), m(t_f)]$ obtained from the proposed method and NLP for cases 1-5 are presented in Table 3. In addition, considering that the NLP solver cannot find convergent solutions for cases 6-10, only the simulation results of LB&TS OCM are presented in Table 4. The terminal states from both methods are obtained by integrating the states from initial point to the terminal time t_f using the obtained control profiles from LB&TS OCM and NLP, respectively. Although the discrete states from the NLP solution yield $x(t_f), y(t_f), z(t_f), v_x(t_f), v_y(t_f), v_z(t_f) = [0, 0, 0, 0, 0, 0]$, special interpolation techniques are required to propagate the solution to the exact terminal point. For a fair comparison, the forward-rectangular integration method is applied to both solutions from NLP and the proposed LB&TS OCM, and the exact terminal states of NLP cannot be realized by directly integrating the dynamics using the discrete control values. As a result, the actual terminal states obtained by integrating the discrete control solutions from NLP generate much larger errors due to discretization. In contrast, with the obtained critical parameters from the well-trained DNNs, LB&TS OCM can find solutions in continuous time. By comparing the terminal states obtained from both methods, results obtained from the proposed method yield smaller errors with much reduced computational time, which demonstrates the computational performance of the proposed LB&TS OCM can find effective control in continuous time. In addition, from the comparison of the final mass, both methods get very similar final mass for cases 1-5, which demonstrates the optimality value obtained from LB&TS OCM. Since a complete optimal solution from the learning-based method is obtained by integrating the system dynamics from an initial point to t_f using the identified critical parameters, the errors at terminal states come from the error of all critical parameters obtained from DNNs. One of the reasons for errors of the critical parameters is due to the limited

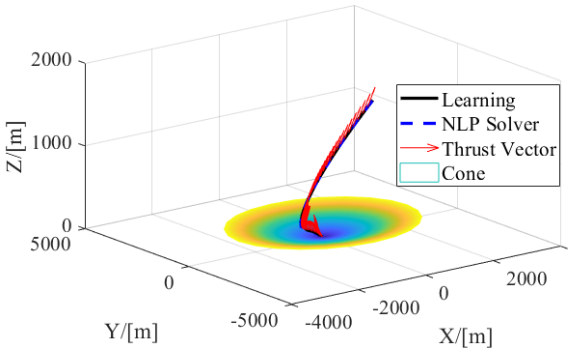
accuracy of the constructed DNNs and the other one is due to the numerical approximation of system dynamics in discretization when generating the dataset of DNNs from NLP.

Table 4 shows that cases 6-10 also have small errors at the terminal point when solved by LB&TS OCM. However, the NLP solver used here cannot find convergent solutions for these cases. These results validate the robustness of the proposed LB&TS OCM. Moreover, Tables 3 and 4 show that the average time for LB&TS OCM to solve the FOPDG problem is less than 0.2 seconds, while the average computational time of NLP is 11.83 seconds. Therefore, we can conclude that the LB&TS OCM is more time efficient than the NLP solver, and the LB&TS OCM has great potential for on-board computation. Besides, results in Tables 3 show that the variance of LB&TS OCM computational time is smaller than of the NLP solver, which indicates predictability of computational time performance when applying the proposed method.

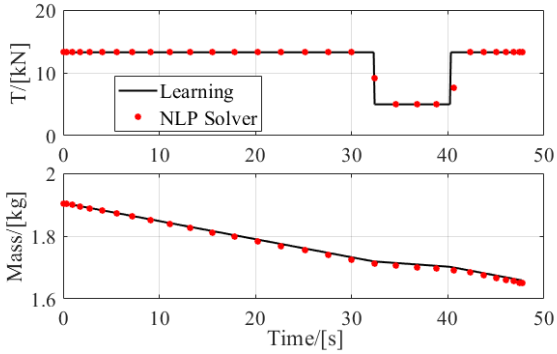
Table 3 Comparison of final states for solving Case 1-5 via LB&TS OCM and NLP solver

		LB&TS OCM		
	$[x, y, z]$/km	$[v_x, v_y, v_z]$/(m/s)	m/kg	Computational time/(s)
$Case1$	$[-0.013, -0.0043, -0.041]$	$[-1.13, 2.63, -1.23]$	1657.5	0.1652
$Case2$	$[0.0021, -0.0044, 0.0015]$	$[-0.24, 0.69, 0.57]$	1634.6	0.1838
$Case3$	$[-0.016, 0.0087, -0.0017]$	$[-0.915, 0.352, 0.682]$	1658.6	0.1717
$Case4$	$[0, 0.032, -0.0085]$	$[0, -1.26, -1.04]$	1734.5	0.1687
$Case5$	$[0.0006, 0.00025, 0.00018]$	$[0.128, -0.047, -0.114]$	1598.9	0.1807
Average	$--$	$--$	$--$	0.1740
		NLP solver		
	$[x, y, z]$/km	$[v_x, v_y, v_z]$/(m/s)	m/kg	Computational time/(s)
$Case1$	$[-0.11, 0.05, -0.101]$	$[-3.3, 10.9, 0.08]$	1657.7	9.67
$Case2$	$[-0.096, 0.002, -0.847]$	$[4.84, -10.4, 0.626]$	1635.9	14.28
$Case3$	$[-0.082, 0.009, -0.11]$	$[-8.67, 7.87, 0.15]$	1659.6	9.70
$Case4$	$[0, 0.058, -0.107]$	$[0, 5.91, -3.36]$	1733.8	9.33
$Case5$	$[0.0005, 0.037, 0.043]$	$[13.6, -5.29, 1.01]$	1597.4	16.19
Average	$--$	$--$	$--$	11.83

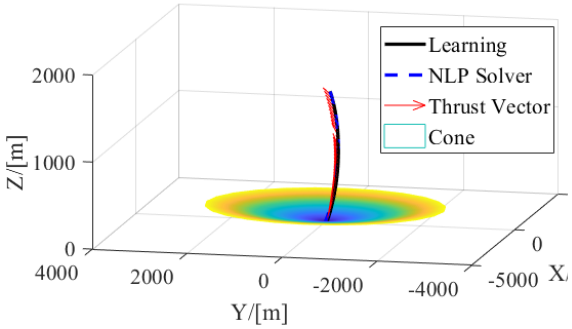
Among the first five cases, trajectories for three of them (cases 3, 4, 5) from both methods are presented in Fig. 9, where the red arrows denote the thrust vectors along the trajectory. For the three presented cases, cases 3 and 5 have only one isolated point while case 4 does not have any isolated point. Furthermore, the time histories of thrust magnitude and mass from both methods for the three cases are almost identical to each other, which demonstrates high accuracy of the training results from DNNs. The time histories of position components $[x, y, z]$, velocity components $[V_x, V_y, V_z]$, and thrust components $[T_x, T_y, T_z]$ for cases 4 and 5 from both methods are presented in Fig. 10. The comparison of control and state components from both methods again demonstrates that the proposed method yields very close results compared to the NLP solutions.



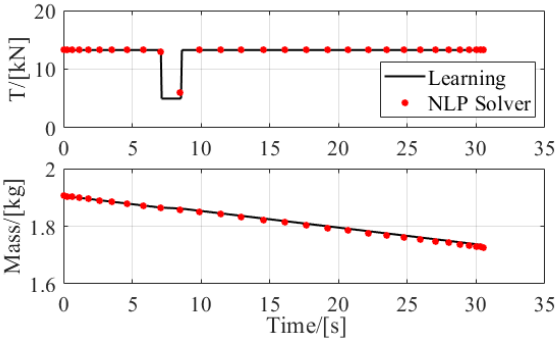
(a) Case 3: The trajectory from two methods



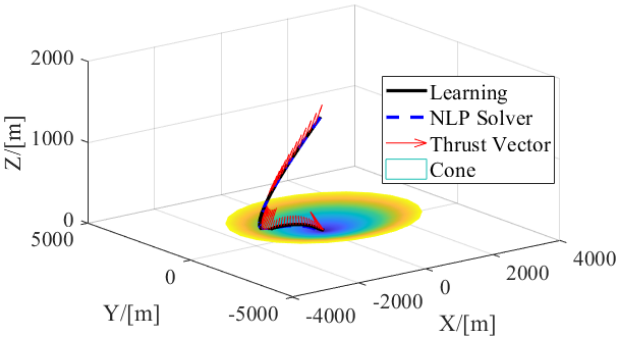
(b) Case 3: The thrust magnitude and vehicle mass from two methods



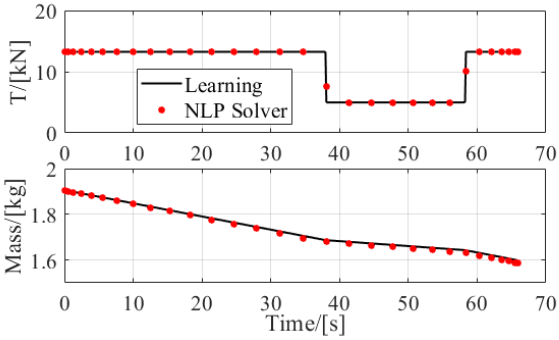
(c) Case 4: The trajectory from two methods



(d) Case 4: The thrust magnitude and vehicle mass from two methods



(e) Case 5: The trajectory from two methods



(f) Case5: The thrust magnitude and vehicle mass from two methods

Fig. 9 The trajectory, thrust magnitude and vehicle mass from learning-based method and NLP

Table 4 The simulation results of Case 6-10 via LB&TS OCM

	LB&TS OCM			
	$[x, y, z]$/km	$[v_x, v_y, v_z]$/(m/s)	m/kg	Computational time/(s)
$Case6$	$[-0.013, -0.013, 0.015]$	$[1.31, 2.86, -2.92]$	1585.1	0.1721
$Case7$	$[-0.010, -0.010, 0.012]$	$[0.95, 0.73, -0.64]$	1672.7	0.1741
$Case8$	$[-0.0098, 0.0020, 0.0078]$	$[-0.915, 0.352, 0.682]$	1659.2	0.1670
$Case9$	$[-0.0052, 0.0033, -0.0016]$	$[0.76, -0.51, -0.53]$	1617.2	0.1687
$Case10$	$[-0.0061, -0.005, -0.0054]$	$[1.00, 0.85, 1.05]$	1643.8	0.1933
Average	$--$	$--$	$--$	0.1750

The state and control variables along the optimal trajectory obtained from the LB&TS OCM are in continuous time. These solutions are obtained without using an on-board optimization solver. Furthermore, with solutions derived from the necessary conditions of optimality, the proposed LB&TS OCM is supported by the optimality theory and can obtain near-optimal solutions. In addition, compared with NLP, the proposed LB&TS OCM based on the refined dataset has advantages in terms of reduced computational time and improved accuracy.

V. Conclusion

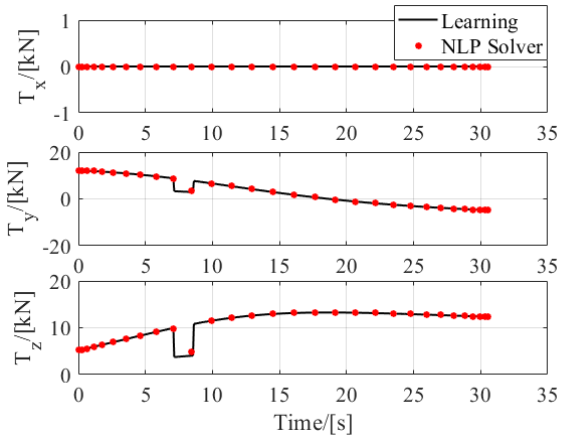
This paper develops a learning-based and theory-supported optimal control method to find the fuel-optimal guidance laws onboard for the three-dimensional powered descent problem. Combined with the Hamiltonian and necessary conditions of optimality, the optimal solution of the fuel-optimal powered descent guidance problem can be represented by a set of identified critical parameters. By generating a sufficient number of cases within a range of problem inputs, the deep neural networks are constructed offline to map the relationship from the problem inputs to the critical parameters. The proposed method requires much less training data compared to the existing learning-based methods, which leads to reduced training time and storage memory. Moreover, the proposed method provides solution in less than 0.2 seconds while reflecting the necessary conditions of optimality derived from the classical Pontryagin's minimum principal. The effectiveness and robustness of the proposed method have been verified using extensive simulation examples.

Acknowledgments

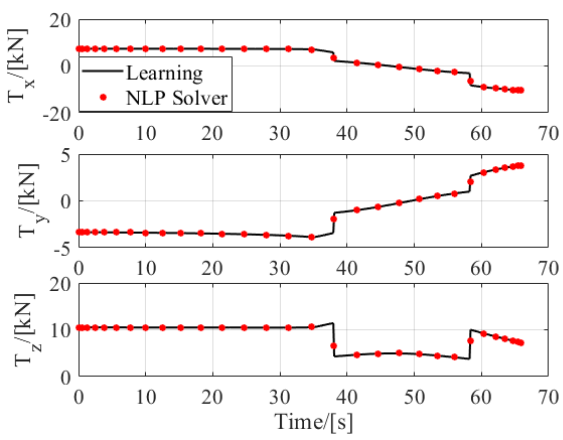
This research has been supported by NASA grant 80NSSC18K1506, and government sponsorship is acknowledged. The authors would like to thank the anonymous reviewers for their suggestions and comments.

References

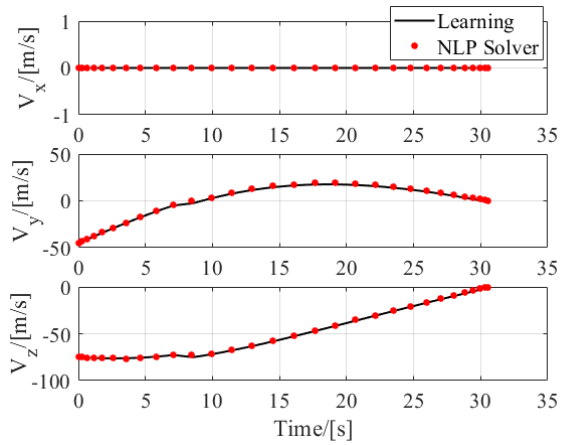
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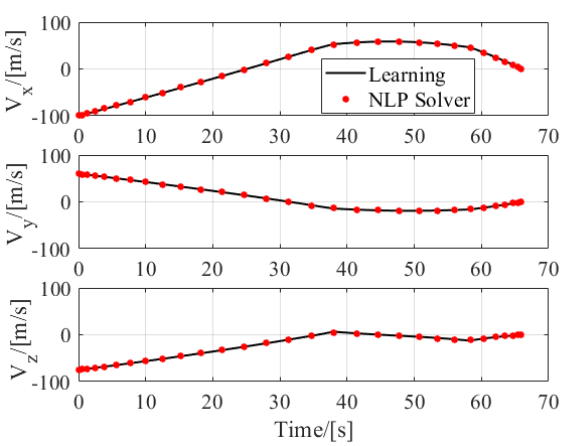
(a) Case 4: The time histories of thrust components from two methods



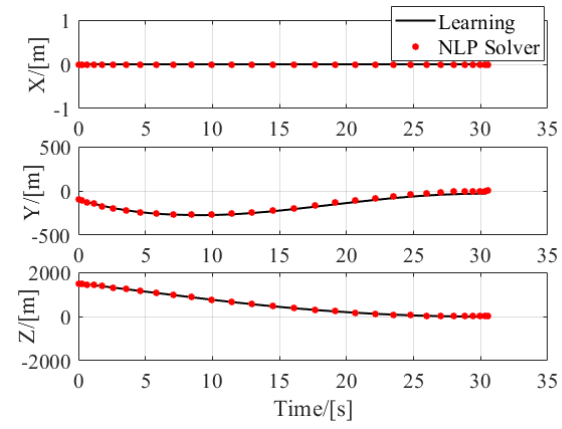
(b) Case 5: The time histories of thrust components from two methods



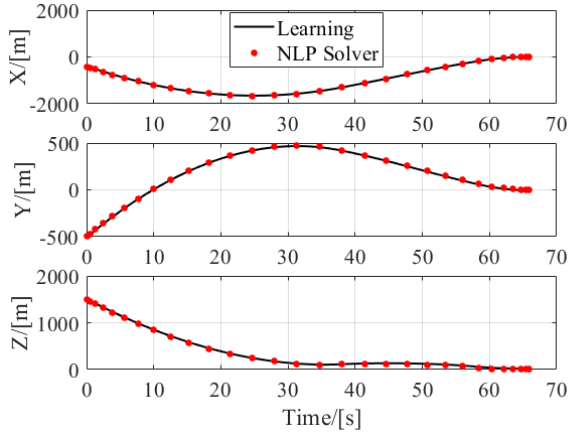
(c) Case 4: The time histories of vehicle speed components from two methods



(d) Case 5: The time histories of vehicle speed components from two methods



(e) Case 4: The time histories of vehicle position components from two methods



(f) Case 5: The time histories of vehicle position components from two methods

Fig. 10 The time histories of position, velocity, and thrust components from the learning-based method and NLP

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